

Multimodal Emotion Interaction and Visualization Platform

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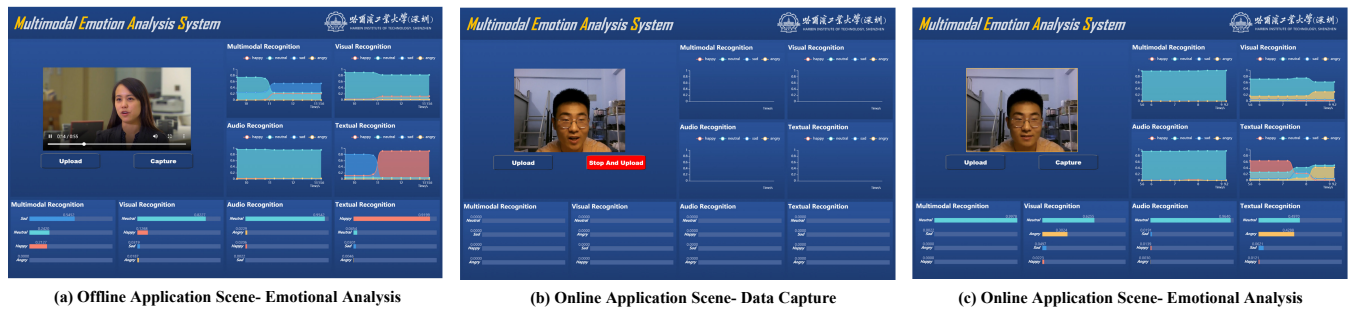


Figure 1: Multimodal Emotion Interaction and Visualization Platform. Our system covers two application scenarios: offline (a) and online (c) application scenes with (b) real-world data collection process.

ABSTRACT

In this paper, we present a multimodal emotion analysis platform, which can flexibly capture, detect and analyze the emotions of video object with multiple modalities under different situations, including offline and online application scenarios. This system can visualize the dynamic effects of different types of emotions from both multimodal and unimodal circumstances. The presented emotion analysis results show instant and time series states in both specific modality and multiple modalities. Our system fills the current research and application gaps in multimodal emotion analysis with an interactive interface. Notably, the constructed system can adaptively process pre-recorded video clips as well as collected real-world data with excellent practicality and interactivity.

CCS CONCEPTS

• **Computing methodologies** → **Artificial intelligence**; • **Human-centered computing** → **Visualization**.

KEYWORDS

Multimodal emotion recognition, human-machine interaction

ACM Reference Format:

Zheng Zhang, Songling Chen, Mixiao Hou, and Guangming Lu. 2023. Multimodal Emotion Interaction and Visualization Platform. In *Proceedings of*

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MM'23, October 2023, Ottawa, Ontario, Canada
© 2023 Copyright held by the owner/author(s).
ACM ISBN 978-8-4007-0108-5/23/10.
<https://doi.org/10.1145/3581783.3612666>

ACM Conference (MM'23). ACM, Ottawa, Ontario, Canada, 3 pages.
<https://doi.org/10.1145/3581783.3612666>

1 INTRODUCTION

Affective computing aims to enable computing machines to perceive, analyze and understand emotions captured from multimodal real-world objectives, which has been recognized as a critical technology for cognitive artificial intelligence. Recent years have witnessed an upsurging number of research on emotion recognition and achieved promising analysis results. The early studies mainly focus on the unimodal emotion analysis on solely visual, audio or text information. Zou et al. [7] designed a co-attention module to detect multi-level acoustic information for speech emotion recognition. Hu et al. [2] proposed a DialogueCRN to proceed with conversational context from a cognitive perspective. Li et al. [3] introduced the DMSRL framework to extract multi-level features during semantic-aware domain adaptation. Although unimodal emotion analysis has made great progress, its recognition performance is ultimately limited by insufficient knowledge. Hence, multimodal emotion analysis appears to achieve promising results by integrating multiple modalities for collective learning. Li et al. [4] provided a decoupled multimodal distillation approach to facilitate flexible and adaptive cross-modal knowledge distillation for enhancing the representation capability of each modality.

Although multimodal emotion recognition has attracted significant enthusiasm in affective computing, it is still at the stage of the theoretical validation level. There are no sufficient mature products for multimodal emotion recognition in the current market. How to build a practical multimodal emotion analysis system and visualize meaningful concrete results is still a demanding task. To this end, this paper develops a multimodal emotion perception, interaction and visualization system to illustrate emotional characters

in multiple video scenes. For enhancing the interactivity of the system, we collect data on the state of the characters in real scenes for dynamic emotion analysis. Moreover, we design two types of data stream visualization results with presentation-instant and time series charts under different modalities, which are designed for long-term mood changes and current and next-point mood prediction.

2 USER INTERFACE

The user interface system shown in Figure 2 includes *i.e.*, video playback area (top left), time series visualization module (top right), and instant visualization module (bottom).

Video Playback Area. This panel allows users to upload objective video clips for emotion analysis with relevant recognition results. There are two buttons ‘Upload’ and ‘Capture’, that is, one for uploading videos (shown in Figure 1 (a) and (c)) and one for capturing material from real-world scenes (shown in Figure 1 (b)).

Instant Visualization Module. The module shows both multimodal and unimodal emotion analysis results. It corresponds to the current emotion moment of the user in the video playback, and the emotion with the highest score will be on top.

Time Series Visualization Module. This module shows the recognition of multimodal and unimodal presentations over a sustained period of time. It can show the emotional changes of the characters to a certain extent.

3 SYSTEM DESIGN

3.1 The Human-Machine Interface

The HMI module is a web application developed based on Electron (a framework for building cross-platform desktop applications using JavaScript, HTML and CSS), Vue.js (a JavaScript framework for building user interfaces), and Axios (a promise-based HTTP library). Moreover, we employ ECharts to present dynamic charts in the interface. The HMI can deliver the video and audio streams from the uploaded video into the data preprocessing module at the back-end service. When acquiring realistic scenes, the HMI will call cameras and recording devices for data acquisition and store them as video files. The visualization module displays the recognition results while playing the video according to the timestamps.

3.2 The Back-end Service

Our back-end services are built on Flask (a lightweight web application framework developed by Python). The back-end service includes two modules, *i.e.*, preprocessing module, and emotional analysis module. The preprocessing module automatically segments the audio stream received from the HMI, and then sends the segmented speech fragments to the speech-to-text module for transcription, followed by multimodal information integration by fusing the three modalities for feature extraction and result analysis.

Segmentation. The data stream slicing is audio-driven, and the audio is sliced into several independent sentences using voice activity detection (VAD) [6]. The timestamps returned by VAD allow the data stream to be split into video and audio segments.

Speech-to-Text. For accessing text, we directly use speech-to-text (STT) [5] to generate transcribed text from audio segments.

Feature Extraction. The visual features are mainly facial clips segmented from video with 5 frames at uniform time intervals by

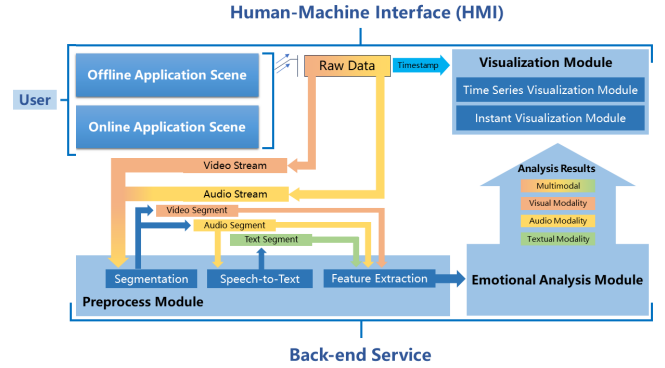


Figure 2: The overall architecture of our system.

facial expression recognition. Textual features are obtained by using the pre-trained Bert model on translated scripts. Audio features are extracted by integrating the log-mel bank features on audio.

Emotional Analysis Module. The multimodal emotion recognition model is built on Pytorch. The model assigns a Bi-GRU to the features of each modality for sequence information learning, except for face images connected to a Resnet-18 network. Then, the sequence features from each modality are semantically learned by the attention mechanism module. Finally, the obtained features are concatenated together and fed to the fully-connected layer for prediction. In the training process, the feature of each modality and the fusion feature are predicted separately for emotion and constrained with a cross-entropy loss function. The unimodal emotion recognition models are the single modal learning branches of the multimodal model. All models are pre-trained on the IEMO-CAP [1], and four categories of samples are reserved, *i.e.*, happy (merged with excited), angry, sad, and neutral. We take the data from the first 4 sessions for training and the last session data for validation to get the optimal model.

4 CONCLUSION

In this article, we develop a multimodal emotional interaction and visualization system. Our system visualizes various application scenarios, including offline and online applications. Notably, both application scenarios can adaptively capture, process and analyze emotions from both single modality and unified multimodal cases. Moreover, the formulated system can dynamic subject detection, automatic model adaption, and visual emotion illustration, with a user-friendly interface.

The system may further involve applications but is not limited to service evaluation (feedback on customer emotions in various business scenarios), teaching evaluation (emotional changes of students in the classroom in response to different teaching content), and security monitoring (fluctuations in the emotional state of characters in specific scenarios). These applications can be built or extended from the current well-designed framework and platform, which will be also be remained in our further work.

Acknowledgment This paper is partially supported by Shenzhen Science and Technology Program under grant RYX20221008092852077, Guangdong Natural Science Foundation under grant 2023A1515010057, and National Key Research and Development Program of China under grant 2018AAA0100100.

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